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Boundary/Topology Posture and Fall-Offs (Pointer)

Assumption A4 is realized either by compact PT-invariant domains with vanishing flux or by standard asymptotically flat/FRW fall-offs; the explicit statements and the symplectic-flux check are compiled in Appendix A (boundary notes therein). We exclude torsion defects and multi-valued ϵ patches that would violate the projector's scalar posture.

Spurion Posture, Ward Identity, and Field Redefinition (Sketch)

Treat ϵ as a spurion transforming as a scalar under diffeos but not varied. Then

$$\delta_\xi S = \int d^4x \sqrt{-g}\, \xi_\nu \Big[\nabla_\mu T^{\mu\nu} + \mathcal{E}^{\mu\nu}_{(e,\omega)} \cdot \delta_\xi(e,\omega) + \nabla_\mu \big(\tfrac{\partial \mathcal{L}}{\partial(\partial_\mu \epsilon)}\, \partial^\nu \epsilon \big) \Big] = 0. \tag{7}$$

Using the Palatini EL equations $\mathcal{E}_{(e,\omega)} = 0$ and C1 ($T_\mu = 3\eta\, \partial_\mu \epsilon$), the spurion current becomes an improvement, hence $\nabla_\mu T^{\mu\nu} = 0$ in the bulk (A4). For Dirac matter, the trace coupling $T_\mu J^\mu$ is removed by $\psi \to e^{i\alpha\epsilon}\psi$; the Jacobian shifts the Nieh–Yan boundary term (A5) but does not affect bulk equations (cf. Section 8).

Summary of Section II. We fixed the global posture (A1–A6), defined the scalar PT projector with a checkable commutation lemma (A3), clarified the PT parities of basic objects and of the Levi–Civita density (A2), and stated conditional selection rules: $\Pi_{\rm PT}$ removes PT-odd scalars and keeps PT-even ones (taking the real part). We emphasized the spurion posture and provided the admissible variational domain that makes the equal–coefficient identity integrable. A compact notation is set in which Σ_ϵ is *by definition* a projected, real scalar and the sign-compensated invariant $\mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \sigma_\epsilon I_T = -6\eta^2|\Sigma_\epsilon|$ will be used uniformly later. These ingredients feed directly into the Palatini analysis (Section III), the uniqueness theorem (Section IV), and the three-chain equivalence (Section V).

3. Palatini Setting and the Allowed PT-Even Scalar Sector

We work in the Palatini posture with independent vierbein $e^A{}_\mu$ and spin connection $\omega^{AB}{}_\mu$ (metric compatible), and we impose the *scalar PT* projector $\Pi_{\rm PT}[\cdot]$ on all observable scalar densities (Section 2). Throughout, Assumptions A1–A5 and the selection rules of Theorem 1 are in force. In particular, all statements below are understood *after* applying $\Pi_{\rm PT}[\cdot]$, so the projected scalars are real.

3.1. Normalized Spurion Direction and Canonical Rank-One Traceless Tensor

The spurion enters the observed sector only through its gradient. We set, once and for all,

$$\Sigma_\epsilon \equiv \Pi_{\rm PT}\Big[(\partial\epsilon)^2\Big], \qquad \mathrm{sgn}(\Sigma_\epsilon) \equiv \mathrm{sgn}(\Sigma_\epsilon), \qquad \hat{n}_\mu \equiv \frac{\partial_\mu \epsilon}{\sqrt{|\Sigma_\epsilon|}},$$

and define the canonical, *dimensionless*, traceless rank-one tensor built from $\hat{n}$,

$$\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu \hat{n}_\nu - \tfrac{1}{4}\delta^\mu{}_\nu\Big), \qquad \mathrm{Tr}\mathcal{T} = 0, \quad \mathrm{Tr}(\mathcal{T}^2) = \tfrac{3}{4}. \tag{8}$$

The (dimension-one) trace-torsion scale along $\hat{n}$ is

$$\tau \equiv \hat{n}^\mu T_\mu \;=\; 3\,\eta\; \mathrm{sgn}(\Sigma_\epsilon)\, \sqrt{|\Sigma_\epsilon|}, \tag{9}$$

which will serve as the unique dimensionful spurion scale at this order.[1]

[1] When $\Sigma_\epsilon = 0$ the constructions are understood on patches with $\Sigma_\epsilon \neq 0$ and extended by continuity. The overall sign bookkeeping is carried by $\mathrm{sgn}(\Sigma_\epsilon)$, so orientation flips are harmless.

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